Chapter 8A Independent sets, coverings and matchings

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8A.1 Introduction

The concepts of independent sets and coverings are used in modeling several real world problems. One of the real world problems called the *Job Assignment Problem* has motivated much research on matchings.

Job-Assignment-Problem (JAP)
There are s persons and t jobs. Each person is capable of handling certain jobs. Under what conditions we can employ **each** of the p persons with a job he/she is capable of handling? The rule of **one-person-one-job** is assumed.

- Notice that some jobs may be left undone.

- Model the problem using appropriate graph theoretic terminology.
- Another popular version of the JAP is known as ***The Marriage Problem***. There are s girls and t boys. Each girl has a list of favorite boys. Under what conditions one can marry off each of the girls to a boy in her list?
- A pure set-theoretic version of JAP is known as ***The Problem of Set Representatives***. Given a set X and a family $F = (X_1, X_2, \ldots, X_s)$ of subsets of X, find necessary and sufficient conditions for choosing s distinct elements $(x_1, x_2, \ldots, x_s)$, such that $x_i \in X_i$, $i = 1, 2, \ldots, s$. The element x_i is called the ***representative*** of X_i.

Notice that if X, F, X_i and s are infinite, then it is a variation of "Axiom of Choice".

8A.2 Independent sets and coverings

We start with a series of definitions.

Definitions. *Let G be a graph.*

- *A set of vertices I is called an **independent set** if no two vertices in I are adjacent. An independent set is also called a **stable set**.*
  - Any singleton set is an independent set. So one is interested to find a largest independent set.
- *The parameter $\alpha_0(G) = \max\{|I| : I \text{ is an independent set in } G\}$ is called the **vertex-independence number** of G.*
- *Any independent set I with $|I| = \alpha_0(G)$ is called a **maximum independent set**.*
- *An independent set I is called a **maximal independent set** if there is no independent set which properly contains I.*
  - Clearly, any maximum independent set is a maximal independent set but a maximal independent set need not be a maximum independent set. An illustration is given below.
- *If $e(u, v)$ is an edge in G, then e is said to **cover** u and v and vice versa.*

- *A set of vertices K is called a **vertex-cover** of G if every edge in G is covered by a vertex in K. That is, if every edge in G has at least one of its end vertices in K.*
  - Clearly, $V(G)$ is a vertex-cover of G. So one is interested to find a smallest vertex cover.
- *The parameter $\beta_0(G) = min\{|K| : K \text{ is a vertex cover of } G\}$ is called the **vertex-covering number** of G.*
- *Any vertex-cover K with $|K| = \beta_0(G)$ is called a **minimum vertex-cover**.*
- *A vertex-cover K is called a **minimal vertex-cover** if there is no vertex-cover which is properly contained in K.*
  - Clearly, every minimum vertex-cover is a minimal vertex-cover but a minimal vertex-cover need not be a minimum vertex-cover. An illustration is given below.

An illustration: In the graph shown in Figure 8A.1, $\{1, 2, 4\}$, $\{3, 5, 6\}$ are maximal independent sets, but they are not maximum independent sets; $\{1, 2, 5, 6\}$ is the maximum independent set. The vertex subset $\{1, 2, 4\}$ is a minimal vertex-cover but it is not a minimum vertex-cover; {3,4} is a minimum vertex-cover.

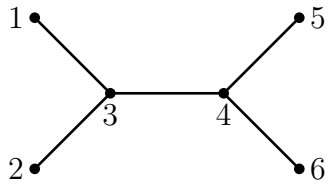


Figure 8A.1: An example to illustrate the independent sets and vertex covers.

This example illustrates that a graph may contain many maximal and maximum independent sets. Similarly, it may contain many minimal and minimum vertex-covers. However, $\alpha_0(G)$ and $\beta_0(G)$ are unique.

The following table shows the independence number and vertex-covering number of some standard graphs.

	P_n	C_n	K_n	K_n^c	$K_{r,s}$
α_0	$\lceil \frac{n}{2} \rceil$	$\lfloor \frac{n}{2} \rfloor$	1	n	$\max\{r, s\}$
β_0	$\lfloor \frac{n}{2} \rfloor$	$\lceil \frac{n}{2} \rceil$	$n-1$	0	$\min\{r, s\}$

Table 8A.1: Table of independence numbers and covering numbers.

The reader may notice that for every graph G shown in the above table, $\alpha_0(G) + \beta_0(G) = |V(G)|$. In fact, this equation holds for any arbitrary graph G. Before proving this claim, we observe a stronger statement.

Theorem 8A.1. *I is an independent set in G iff $V(G) - I$ is a vertex-cover of G.*

Proof. I is an independent set

$\Leftrightarrow$ no two vertices in I are adjacent

$\Leftrightarrow$ no edge has both of its end vertices in I

$\Leftrightarrow$ every edge in G has an end vertex in $V(G) - I$

$\Leftrightarrow$ $V(G) - I$ is a vertex-cover of G. □

Corollary. *For any graph G, $\alpha_0(G) + \beta_0(G) = n(G)$.*

Proof. Let I be a maximum independent set. By the above theorem, $V(G) - I$ is a vertex cover of G. Hence, $\beta_0(G) \leq |V(G) - I| = n - \alpha_0(G)$; that is $\alpha_0(G) + \beta_0(G) \leq n$.

Let K be a minimum vertex-cover of G. By the above theorem, $V(G) - K$ is an independent set of vertices. Hence, $\alpha_0(G) \geq |V(G) - K| = n(G) - \beta_0(G)$; that is $\alpha_0(G) + \beta_0(G) \geq n(G)$.

The two inequalities imply the corollary. □

We now define the edge analogues of independent sets of vertices and vertex-covers.

Definitions.

- *A subset of edges M in G is called an **independent set** of edges if no two edges in M are adjacent. An independent set of edges is more often called as a **matching.***
  - Clearly, if M is a singleton, then it is a matching. So one is interested to find a largest matching.
- *The parameter $\alpha_1(G) = max\{|M| : M$ is a matching in $G\}$ is called the **matching number** of G.*
- *Any matching M with $|M| = \alpha_1(G)$ is called a **maximum matching**.*
- *A matching M is called a **maximal matching** if there is no matching which properly contains M.*
- *A set of edges F is called an **edge-cover** of G if every vertex in G is incident with an edge in G.*
  - A graph G need not have an edge cover; for example, $K_2 \cup K_1$ has no edge-cover. In fact, a graph G has an edge cover iff $\delta(G) > 0$.
  - If $\delta(G) > 0$, then $E(G)$ is an edge-cover. So one is interested to find a smallest edge-cover.
- *If $\delta(G) > 0$, then the parameter $\beta_1(G) = min\{|F| : F$ is an edge cover of $G\}$ is called the **edge covering number** of G.*
- *Any edge-cover F with $|F| = \beta_1(G)$ is called a **minimum edge cover.***
- *An edge-cover F is called a **minimal edge-cover** if there is no edge-cover which is properly contained in F.*

An illustration: In the graph of Figure 8A.2, $\{a, d\}$ and $\{b, c\}$ are maximal matchings but they are not maximum matchings ; $\{a, e, f\}$ is a maximum matching. $\{b, c, e, f\}$ is a minimal edge-cover but it is not a minimum edge-cover. $\{a, e, f\}$ is a minimum edge-cover.

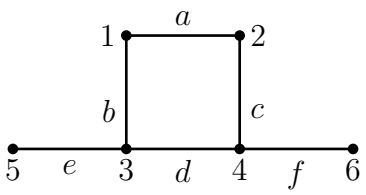


Figure 8A.2: An example to illustrate matchings and edge covers.

Again we observe that a graph G may contain many maximal and maximum matchings. It also may contain many minimal and minimum edge covers. But $\alpha_1(G)$ and $\beta_1(G)$ are unique.

The following table shows the matching-number and edge-covering number of some standard graphs.

	P_n	C_n	K_n	$K_{r,s}$
α_1	$\lfloor \frac{n}{2} \rfloor$	$\lfloor \frac{n}{2} \rfloor$	$\lfloor \frac{n}{2} \rfloor$	$\min\{r, s\}$
β_1	$\lceil \frac{n}{2} \rceil$	$\lceil \frac{n}{2} \rceil$	$\lceil \frac{n}{2} \rceil$	$\max\{r, s\}$

Table 8A.2: Matching numbers and edge-covering numbers.

Again notice that for every graph G shown in the above table $\alpha_1(G) + \beta_1(G) = n(G)$. However, the edge analogue of Theorem 8A.1 does not hold. That is, if M is a matching in G, then $E(G) - M$ need not be an edge cover of G. And if F is an

edge cover of G, then $E(G) - F$ need not be a matching. Reader is encouraged to construct graphs to justify these claims. However, $\alpha_1(G) + \beta_1(G) = n(G)$, for any graph G with $\delta(G) > 0$. Before proving it we introduce a new concept.

Definitions.

- *Let M be a matching in G. A vertex v is said to be* ***M-saturated*** *if there exists an edge in M which is incident to v; else v is said to be* ***M-unsaturated.***
- *A matching M is said to be a* ***perfect matching*** *if it saturates every vertex of G.*
- *An M-alternating path is a sequence of distinct vertices $\langle v_1.v_2, \ldots, v_k\rangle$ with $k \geq 2$ where v_1, v_k are M-unsaturated and $(v_i, v_{i+1}) \in M$ for every even index i and $(v_i, v_{i+1}) \in E(G) \setminus M$ for every odd index i.*

In the following graph, $M_1 = \{(1,2),(4,5)\}$ is a matching. The vertices 1, 2, 4 and 5 are M_1-saturated but 3 and 6 are M_1-unsaturated. So, it is not a perfect matching. While the path $\langle 3,4,5,6\rangle$ is an M_1-alternating path connecting the unsaturated vertices 3 and 6. . However, $M_2 = \{(1,2),(3,4),(5,6)\}$ is a perfect matching.

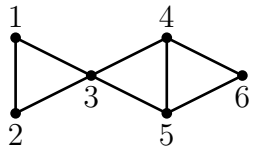


Figure 8A.3: A graph to illustrate perfect matching.

Remarks.

- Every perfect matching is a maximum matching. However, the converse is false.
- If G has perfect matching, then $n(G)$ is even. However, the converse is false; for example, $K_{1,3}$ and $K_{2,4}$ do not have perfect matchings.

Theorem 8A.2. *For any graph G with $\delta(G) > 0$, $\alpha_1(G) + \beta_1(G) = n(G)$.*

Proof. It consists of two steps.

(1) $\alpha_1(G) + \beta_1(G) \leq n(G)$.

Let M be a maximum matching and U be the set of all M-unsaturated vertices. Then $|U| = n - 2|M|$. Let $U = \{v_1, v_2, \ldots, v_p\}$, where $p = n - 2|M|$. Let e_i be an edge incident to v_i, $i = 1, \ldots, p$; such an edge exists, since $\delta(G) > 0$. Then $M \cup \{e_1, e_2, \ldots, e_p\}$ is an edge-cover. So,

$$\beta_1(G) \leq |M| + p \leq |M| + (n - 2|M|) = n - |M| = n - \alpha_1(G).$$

Hence, $\alpha_1(G) + \beta_1(G) \leq n$.

(2) $\alpha_1(G) + \beta_1(G) \geq n(G)$.

Let F be a minimum edge cover. Let H be the spanning subgraph of G with edge set F. Let M_H be a maximum matching in H and U be the set of M_H-unsaturated vertices in H. As before, $|U| = n - 2|M_H|$. Let $U = \{v_1, v_2, \ldots, v_p\}$, where $p = n - 2|M_H|$. Since, F is an edge-cover, it contains an edge e_i incident with v_i, $1 \leq i \leq p$. Since, M_H is a maximum matching, U is an independent set in H. Therefore $e_i \neq e_j$, if $v_i \neq v_j$. Hence, $e_1, e_2, \ldots, e_p$ are all distinct edges. It follows that $|F| \geq |M_H| + p = |M_H| + n - 2|M_H| = n - |M_H|$. Hence, $\beta_1(G) = |F| \geq n - |M_H| \geq n - \alpha_1(G)$, since $\alpha_1(G) \geq |M_H|$. So, $\alpha_1(G) + \beta_1(G) \geq n$.

The two inequalities imply the theorem. □

8A.3 Matchings in bipartite graphs

We have now enough terminology to model the job-assignment-problem and solve it.

Graph theoretic model: Let $P_1, P_2, \ldots, P_s$ be the s persons and $J_1, J_2, \ldots, J_t$ be the t jobs. Represent each P_i by a vertex p_i and each job J_h by vertex j_h. Join p_i and j_h by an edge if P_i is capable of handling j_h. This construction yields a bipartite graph $G[X, Y]$, where $X = \{p_1, \ldots, p_s\}$ and $Y = \{j_1, \ldots, j_t\}$. The problem is to find necessary and sufficient conditions for G to contain a matching saturating every vertex in X. The following example illustrates the construction of G.

Persons	Jobs P_i can handle
P_1	J_1, J_3
P_2	J_1, J_2, J_3
P_3	J_2, J_4, J_5
P_4	J_3

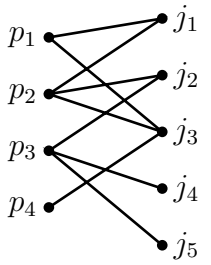


Figure 8A.4: A job assignment as required is possible in this example. One possible assignment is to assign P_1 to J_1, P_2 to J_2 P_3 to J_5 and P_4 to J_3.

If G$[X, Y]$ contains a matching, say $\{(p_1, j_1), \ldots, (p_s, j_s)\}$ saturating every vertex in X, then p_i can be assigned the job j_i, $1 \leq i \leq s$.

A necessary condition for the existence of a solution for JAP is intuitively obvious: Every set of k persons ($1 \leq k \leq s$) must be able to handle at least k jobs together. That is, in graph theoretic terminology, if G has a matching saturating every vertex in X, then every k vertices in X are adjacent to at least k vertices in Y ($1 \leq k \leq s$). Hall (1939) showed that the converse is also true.

Notation: If $G[X, Y]$ is a bipartite graph and $S \subseteq X$, let $N_G(S) = \{y \in Y : y \text{ is adjacent to some vertex in } S\}$.

• Hall's Theorem

Theorem 8A.3 (Hall, 1939)**.** *A bipartite graph $G[X, Y]$ has a matching saturating every vertex in X iff*

(Hall) $\quad |N_G(S)| \geq |S|$, *for every* $S \subseteq X$.

Proof. Let $X = \{x_1, x_2, \ldots, x_s\}$ and $Y = \{y_1, y_2, \ldots, y_t\}$.

Necessity ($\Rightarrow$)**:** Suppose G has a matching $M = \{(x_1, y_1), \ldots, (x_s, y_s)\}$ saturating every vertex in X. Let $S = \{x_1', x_2', \ldots, x_r'\} \subseteq X$. Then $y_1', y_2', \ldots, y_r' \in N_G(S)$ and they are all distinct, since M is a matching. So $|N_G(S)| \geq r = |S|$.

Sufficiency ($\Leftarrow$)**:** There are many proofs of sufficiency. The following proof is due to Halmos and Vaughan (1950). It is by induction on s.

If $s = 1$, then the result is obvious. So, let $s \geq 2$ and proceed to the next step in the induction. We make two cases.

Case 1: For every proper subset S of X, $|N_G(S)| \geq |S| + 1$.

Since G satisfies (Hall), $|N(\{x_1\})| \geq 1$; and so there is a vertex, say $y_1 \in Y$ such that (x_1, y_1) is an edge. Consider the subgraph $L = G - \{x_1, y_1\}$ whose bipartition is $[X_1, Y_1]$, where $X_1 = \{x_2, \ldots, x_s\}$ and $Y_1 = \{y_2, \ldots, y_t\}$. We claim that $L(X_1, Y_1)$ satisfies (Hall). So, let $T \subseteq X_1$. Clearly, $N_G(T) \subseteq N_L(T) \cup \{y_1\}$. Hence,

$$\begin{aligned} |N_L(T)| &\geq |N_G(T)| - 1, \\ &\geq (|T| + 1) - 1, \text{by our assumption}, \\ &= |T|. \end{aligned}$$

Therefore, by induction hypothesis, $L[X_1, Y_1]$ contains a matching M saturating every vertex in X_1. Then $M \cup \{(x_1, y_1)\}$ is a required matching of G.

Case 2: For some proper subset S of X, $|N_G(S)| = |S|$.

For notational convenience, let $S = \{x_1, x_2, \ldots, x_p\}$ and $N(S) = \{y_1, y_2, \ldots, y_p\}$.

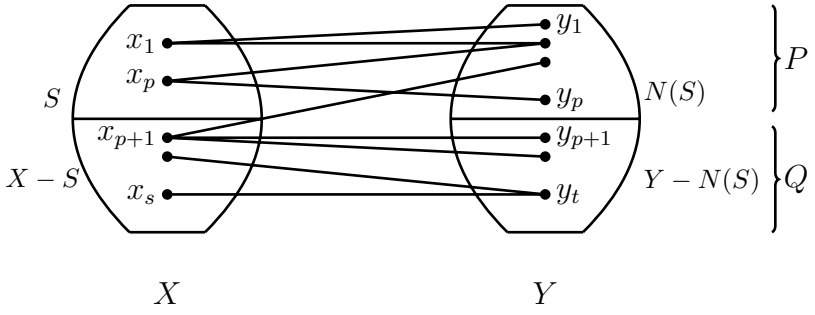


Figure 8A.5: The bipartite graphs $P[S, N(S)]$ and $Q[X - S, Y - N(S)]$.

Consider the bipartite subgraphs $P[S, N(S)]$ and $Q[X - S, Y - N(S)]$.

We shall prove that both these graphs satisfy (Hall) and then complete the proof by appealing to the induction hypothesis.

We consider first $P[S, N(S)]$. Let $T \subseteq S$. Clearly, by our assumption of S, $N_P(T) = N_G(T)$; so $|N_P(T)| \geq |T|$. Thus P satisfies (Hall), and hence, by induction hypothesis, P has a matching, say M_1 saturating every vertex in S. Next, consider $Q[X - S, Y - N(S)]$. Let $T \subseteq X - S$. Clearly, $N_G(S \cup T) \subseteq N_G(S) \cup N_Q(T)$. So,

$$\begin{aligned} |N_Q(T)| &\geq |N_G(S \cup T)| - |N_G(S)|, \\ &\geq |S \cup T| - |S|, \text{ since G satisfies (Hall)}, \\ &= |S| + |T| - |S|, \text{ since } S \text{ and } T \text{ are disjoint}, \\ &= |T|. \end{aligned}$$

Thus $Q[X - S, Y - N_G(S)]$ satisfies (Hall). So by induction hypothesis, Q has a matching, say M_2 saturating every vertex in $X - S$. Then $M_1 \cup M_2$ is a required matching. □

Two interesting corollaries follow.

Corollary. *Let G be a bipartite graph with bipartition $[X, Y]$ and $\delta(G) > 0$. If*

$$\min_{x \in X} deg(x) \geq \max_{y \in Y} deg(y),$$

then G contains a matching saturating every vertex in X.

Proof. It is enough if we prove that G satisfies (Hall). So, let $S \subseteq X$. Let $\delta(X) = \min_{x \in X} deg(x)$ and $\Delta(Y) = \max_{y \in Y} deg(y)$. If e is an edge incident with a vertex in X, then its other end vertex is in $N(S)$. So, there are at least $|S| \cdot \delta(X)$ edges incident with the vertices of $N(S)$. Hence, there is a vertex $y \in N(S)$ which is incident with at least $\frac{|S| \cdot \delta(X)}{|N(S)|}$ edges (: Use the pigeon-hole principle). So, $\Delta(Y) \geq deg(y) \geq \frac{|S| \cdot \delta(X)}{|N(S)|}$. Since, $\delta(X) \geq \Delta(Y)$, we obtain $|N(S)| \geq |S|$. And thus, we have verified (Hall). □

Corollary. *Let $G[X, Y]$ be a k-regular ($k \geq 1$) bipartite graph. Then $E(G)$ can be partitioned into k sets $E_1, E_2, \ldots, E_k$, where each E_j is a perfect matching.*

Proof. We shall prove the theorem by induction on k. At the outset observe that $|X| = |Y|$, since $k|X| = |E(G)| = k|Y|$. If $k = 1$, then $E(G)$ is a perfect matching and hence the assertion follows. So we proceed to the next step in the induction.

Since, $G[X, Y]$ is a k-regular bipartite graph it satisfies the hypothesis of the above corollary. Hence, it contains a perfect matching, say E_1. Since, $|X| = |Y|$, $G - E_1$ is a $(k-1)$-regular bipartite graph. Therefore, by induction hypothesis, $E(G - E_1)$ can be partitioned into $k - 1$ sets, say $E_2, \ldots, E_k$, where each E_i is a perfect matching. Then $(E_1, E_2, \ldots, E_k)$ is a required partition of $E(G)$. □

• König's Theorem

There are several theorems in discrete mathematics which show that a minimum parameter is equal to a maximum parameter. The next result is one such

theorem. Before stating it, we observe an important inequality. If M is a matching and K is a vertex-cover of G, then any vertex of K covers at most one edge of M. Hence:

$$\boxed{\beta_0(G) \geq \alpha_1(G), \text{ for any graph } G.}$$

Remarks.

- $\beta_0(C_{2k}) = \alpha_1(C_{2k}) = k$.
- $\beta_0(K_{p,p}) = \alpha_1(K_{p,p}) = p$.
- $\beta_0(C_{2k+1}) = k+1$, $\alpha_1(C_{2k+1}) = k$.

The above remarks lead to the following unsolved question.

$$\boxed{\text{For which graphs } G,\ \beta_0(G) = \alpha_1(G)?}$$

The following result identifies one such class of graphs.

Theorem 8A.4 (König, 1931)**.** *For any bipartite graph $G[X, Y]$,*

$$\beta_0(G) = \alpha_1(G).$$

Proof. In view of the above observation, we have to only show that $\alpha_1(G) \geq \beta_0(G)$. Let $K = A \cup B$ be a minimum vertex-cover, where $A \subseteq X$ and $B \subseteq Y$. So, $|A| + |B| = |K| = \beta_0(G)$. We consider the bipartite subgraphs $P[A, Y - B]$ and $Q[B, X - A]$.

Claim 1: $P[A, Y - B]$ has a matching M_1 saturating every vertex in A.

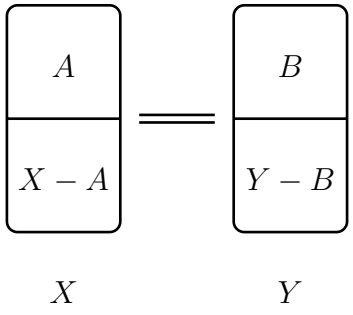


Figure 8A.6: A bipartite graph with vertex-cover $A \cup B$.

To prove this claim it is enough if we verify that P satisfies (Hall). On the contrary, suppose that there exists some $S \subseteq A$ such that $|N_P(S)| < |S|$. Then it is easy to verify that the set $(A - S) \cup N_P(S) \cup B$ is a vertex cover of G. So,

$$\begin{aligned} |(A-S) \cup N_P(S) \cup B| &= |A| - |S| + |N_P(S)| + |B|, \\ &< |A| + |B|, \text{ since by our assumption, } |N_p(S)| < |S|, \\ &= |K|. \end{aligned}$$

This is a contradiction to the minimality of K. Therefore, by Hall's theorem, there exists a matching M_1 in P saturating every vertex in A.

Claim 2: Similarly, it follows that $Q[B, X - A]$ has a matching M_2 saturating every vertex in B.

Clearly $M_1 \cap M_2 = \emptyset$. Hence,

$$\begin{aligned} \alpha_1(G) &\geq |M_1| + |M_2|, \\ &= |A| + |B|, \\ &= |K| = \beta_0(G). \end{aligned}$$

□

• Alternating Paths

Lemma 8A.5. *Let M be a matching in a graph $G = (V; E)$ and P the set of edges of some M-alternating path. Let $M' = (M \setminus P) \cup (P \setminus M)$. Then M' is a matching and $|M'| = |M| + 1$.*

Proof. Let $\langle v_1, v2, \ldots, v_k \rangle$ be the sequence of vertices of the M-alternating path. Note $M \setminus P$ is a matching that leaves all of $v_1, v_2, \ldots, v_k$ unsaturated. Also note that $P \setminus M = \{(v_1, v_2), (v_3, v_4), \ldots, (v_{k-1}, v_k)$ is a matching that only matches vertices on M. Therefore, $M' = (M \setminus P) \cup (P \setminus M)$ is a matching.

Conclude by noting that $|M'| - |M| = |P \setminus M| - |P \cap M| = \frac{k}{2} - (\frac{k}{2} - 1) = 1$. □

Thus, we can find larger matchings by finding alternating paths. It turns out that the lack of an M-alternating path conveys that M is a maximum matching.

Lemma 8A.6. *M is a maximum matching if and only if there is no M-alternating path.*

Proof. We already showed that the existence of an M-alternating path implies M is not a maximum matching. So now suppose M is not a maximum matching. We will find an M-alternating path.

Let M^* be a maximum matching, so $|M*| > |M|$. Construct the graph $\mathcal{H} = (V; M \cup M^*)$, meaning add each edge from M and M^* and keep two copies of each edge $e \in M \cap M^*$ (note $\mathcal{H}$ may have parallel edges). Since M and M^* are matchings, then $deg_{\mathcal{H}}(v) \leq 2$ for each $v \in V$. This is equivalent to saying that $\mathcal{H}$ is comprised of vertex-disjoint paths and cycles (parallel edges are length-2 cycles). Every cycle in $\mathcal{H}$ alternates between edges in M and edges in M^* so $|C \cap M^*| = |C \cap M|$ for every cycle C. This, plus the fact that $|M^*| > |M|$, means there is a path P in

$\mathcal{H}$ with $|P \cap M^*| > |P \cap M|$. Such a path P starts and ends with edges in M^*, so its endpoints are M-unsaturated. That is, P is an M-alternating path. □

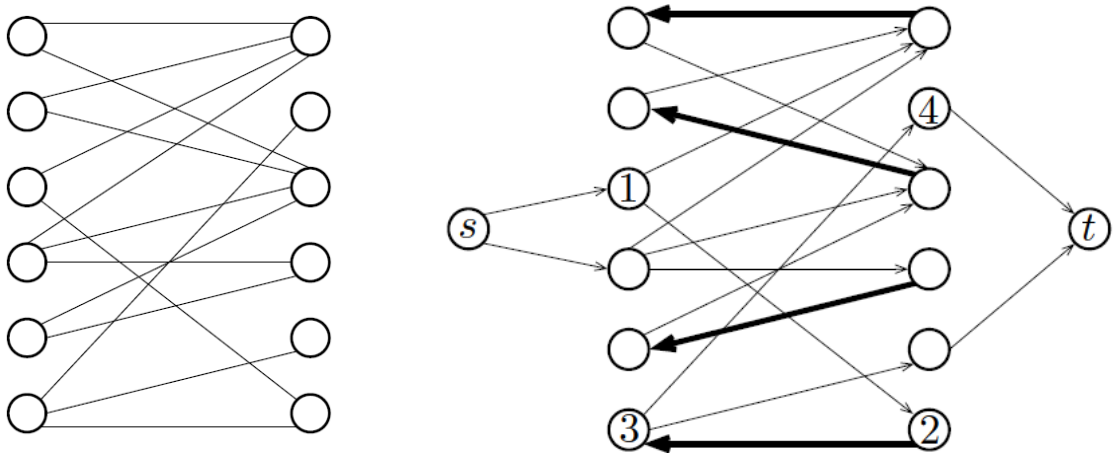
In proving Lemmas 8A.5 and 8A.6, we did not rely on the bipartite structure of G. These will also be the foundation of our Maximum Matching algorithm in general graphs. The main effort is then to find an alternating path. It is much simpler to find such a path in a bipartite graph.

• Finding Alternating Paths in Bipartite Graphs

Lemma 8A.7. *Given a matching M in a bipartite graph $G = (V; E)$ we can either find an M-alternating path or determine that no M-alternating path exists in $O(m)$ time.*

Proof. Let $V = [X \cup Y]$ be the bipartition of G. Let $\overrightarrow{G_M}$ be the directed graph $(V \cup \{s, t\}, \overrightarrow{E_M})$ obtained from G by first directing every edge of M to point toward X and directing each edge of $E \setminus M$ to point toward Y. Then add two new auxiliary vertices s, t to $\overrightarrow{G_M}$; the vertex s has an outgoing edge to each M-unsaturated vertex in X and the vertex t has and incoming edge from each M-unsaturated vertex in Y.

Any $s - t$ path $\overrightarrow{P}$ in $\overrightarrow{G_M}$ corresponds to an M-alternating path: if we remove s, t from $\overrightarrow{P}$ then it starts and ends at M-unsaturated vertices. Furthermore, by how the edges of G were oriented, the (original undirected) edges used by $\overrightarrow{P}$ alternate between an edge of $E \setminus M$ and an edge of M. Conversely, any M-alternating path $\langle v_1, v_2, \ldots, v_k \rangle$ in G can be extended to an $s - t$ path in $\overrightarrow{G_M}$ in the obvious way: In $\overrightarrow{G_M}$ the path is $\langle s, v_1, v_2, \ldots, v_k, t \rangle$. Such an $s - t$ path can be found in $O(m)$ time using, say, a breadth first search. □

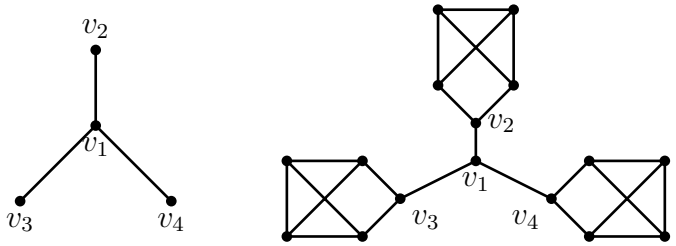


(a) A bipartite graph G (b) Directed graph from a matching (in bold)

Figure 8A.7: An $s-t$ path $\langle s, 1, 2, 3, 4, t\rangle$ in the directed graph and a corresponding to the alternating path $\langle 1, 2, 3, 4\rangle$ in G.

8A.4 Perfect matchings in graphs

All the graphs in this section are simple graphs. If G has a perfect matching, then $n(G)$ is even, obviously. However, the converse does not hold. For example, the two graphs G_1 and G_2 shown in Figure 8A.8 have no perfect matchings; this follows since only one of the vertices v_2 or v_3 or v_4 can be matched with v_1.



(a) G_1 (b) G_2

Figure 8A.8: Graphs with no perfect matchings.

Definition. *A component of G is called an **odd component** if it has odd number of vertices, and it is called an **even component** if it has an even number of vertices. The number of odd components in G is denoted by $\theta(G)$.*

Lemma 8A.8. *If $n(G)$ is even and $S \subseteq V(G)$, then $\theta(G-S)$ and $|S|$ have the same parity.*

Proof. Exercise. □

We can now proceed to state and prove a theorem which characterizes the graphs with perfect matchings.

Theorem 8A.9 (Tutte, 1947)**.** *A graph G has a perfect matching iff*

(Tutte) $\theta(G-S) \leq |S|$, *for every* $S \subseteq V(G)$.

Proof. (1) G has a perfect matching $\Rightarrow$ G satisfies (Tutte).

Let M be a perfect matching in G and let $S = \{v_1, v_2, \ldots, v_s\} \subseteq V(G)$. Let $G_1, \ldots, G_t$ be the odd components of $G-S$, where $t = \theta(G-S)$. Since, G_i $(1 \leq i \leq t)$ has odd number of vertices, at least one vertex u_i of G_i is matched under M with a vertex v_i of S; that is $(u_1, v_1), \ldots, (u_t, v_t) \in M$. Since M is a matching, v_i's are all distinct. Hence, $|S| \geq t = \theta(G-S)$.

(2) G satisfies (Tutte) $\Rightarrow$ G has a perfect matching.

We give a proof that is due to Anderson (1971). It is by induction on n. We first observe that n is even: By taking $S = \emptyset$ in (Tutte) we deduce that $\theta(G) = \theta(G-\emptyset) \leq |\emptyset| = 0$. That is, G has no odd components. Hence, n is even. We now proceed to prove the implication.

Basic step $n = 2$**:** Since G satisfies (Tutte), $G \simeq K_2$ and it has a perfect matching.

Induction step: We make two cases.

Case 1: $\theta(G-S) \leq |S| - 1$, for every proper subset $S \subseteq V(G)$.

By Lemma 8A.8, $\theta(G-S) \neq |S|-1$, for any S. So,

(P_1) $\theta(G-S) \leq |S|-2$, for every proper subset $S \subseteq V(G)$.

Let $e=(u,v)$ be an edge in G, and let $H = G - \{u,v\}$.

Claim: H satisfies (Tutte).

On the contrary, suppose that H contains a $T \subseteq V(H)$ such that $\theta(H-T) \geq |T|+1$.

Since, $n(H)$ is even, using Lemma 8A.8, we conclude that $\theta(H-T) \geq |T|+2$. But then

$$\theta(G-(T\cup\{u,v\})) = \theta(H-T) \geq |T|+2 = |T\cup\{u,v\}|.$$

Since G satisfies (Tutte), $\theta(G-(T\cup\{u,v\})) \leq |T\cup\{u,v\}|$. Hence, $\theta(G-(T\cup\{u,v\})) = |T\cup\{u,v\}|$. This contradicts our assumption. Hence, the claim holds. Therefore, by induction hypothesis, H has a perfect matching, say M. But then $M\cup\{e\}$ is a perfect matching of G.

Case 2: $\theta(G-S) = |S|$, for some proper subset $S \subseteq V(G)$.

Let $S^* = \{v_1, v_2, \ldots, v_s\}$ be a subset with ***maximum number*** of vertices such that $\theta(G-S^*) = |S^*|$. We look at $G-S^*$. Let $G_1, \ldots, G_s$ be the odd components of $G-S^*$.

Observation 1: $G-S^*$ has no even components.

On the contrary, if D is an even component of $G-S^*$ and $x \in V(D)$, then $D-x$ has at least one odd component and so $\theta(G-(S^*\cup\{x\})) \geq s+1$. Hence, $\theta(G-(S^*\cup\{x\})) = |S^*\cup\{x\}|$, since G satisfies (Tutte). This is a contradiction to the maximality of $|S^*|$.

Observation 2: Any k $(1 \leq k \leq s)$ odd components from $G_1, \ldots, G_s$ are together joined to at least k vertices of S^*; else, G violates (Tutte).

Therefore, by Hall's Theorem 8A.3, there exists a matching, say $M_0 = \{(u_1, v_1), \ldots, (u_s, v_s)\}$, where $u_i \in G_i$ $(1 \leq i \leq s)$.

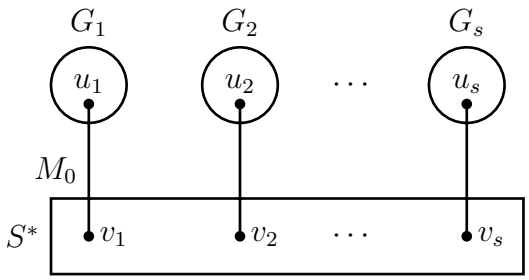


Figure 8A.9: The graph G with matching $M_0 = \{(u_1, v_1), \ldots, (u_s, v_s)\}$.

Figure 8A.9 shows the structure of G that we have derived until now.

Observation 3: $H_1 = G_1 - u_1$ satisfies (Tutte).

On the contrary, suppose that there exists $T \subseteq V(H_1)$ such that $\theta(H_1 - T) \geq |T|+1$. Since, $n(H_1)$ is even, using Lemma 8A.8, we conclude that $\theta(H_1 - T) \geq |T|+2$. Let $L_1, L_2, \ldots, L_p$ be the odd component of $H_1 - T$, where $p \geq |T| + 2$. But then $L_1, L_2, \ldots, L_p$ and $G_2, \ldots, G_s$ are odd components of $G - (S^* \cup \{u_1\} \cup T)$. Hence,

$$\begin{aligned} \theta(G - (S^* \cup \{u_1\} \cup T)) &= p + (s - 1), \\ &\geq |T| + 2 + s - 1, \\ &= |T| + 1 + s, \\ &= |(S^* \cup \{u_1\} \cup T)|. \end{aligned}$$

Since G satisfies (Tutte), we conclude that $\theta(G - (S^* \cup \{u_1\} \cup T)) = |S^* \cup \{u_1\} \cup T|$, which is a contradiction to the maximality of $|S^*|$. Hence the observation is proved.

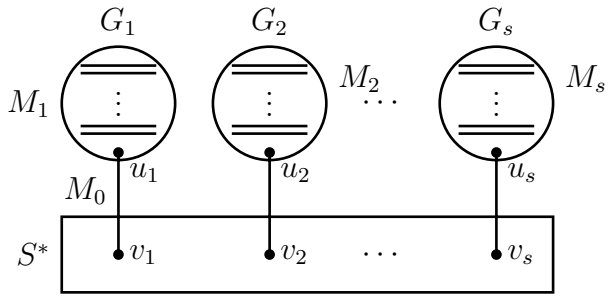


Figure 8A.10: G with perfect matchings $M_0 \cup M_1 \cup \ldots \cup M_s$.

Therefore, by induction hypothesis, H_1 has a perfect matching, say M_1. Similarly, $G_2 - \{u_2\}, \ldots, G_s - \{u_s\}$ contain perfect matchings, say $M_2, \ldots, M_s$ respectively. But then $M_1 \cup M_2 \cup \cdots \cup M_s$ is a perfect matching of G; see Figure 8A.10. □

Although Tutte's theorem characterizes the graphs with a perfect matching, it is hard to verify Tutte's condition and conclude that a given graph G has a perfect matching, because we have to verify (Tutte) for 2^n subsets of $V(G)$. Hence, there have been several results proved by various mathematicians which say that a given graph G has a perfect matching if G satisfies a certain property (P)(where (P) is easily verifiable). In fact, the first result on perfect matchings was obtained by Petersen (1891) which preceded Tutte's theorem. However, we can easily deduce Petersen's result using Tutte's theorem.

Corollary. *If G is a $(k-1)$-edge-connected k-regular graph with n even, then G has a perfect matching.*

Proof. It is enough if we verify that G satisfies (Tutte). Let $S \subseteq V(G)$ and $G_1, G_2, \ldots, G_t$ be the odd components of $G - S$. Since, G is $(k-1)$-edge-connected, $|[S, V(G_i)]| \geq k-1$, for every $i, 1 \leq i \leq t$.

Claim: $|[S, V(G_i)]| \geq k$, for every $i, 1 \leq i \leq t$.

On the contrary, suppose that for some j, $1 \leq j \leq t$, $|[S, V(G_j)]| = k - 1$. Then

$$\begin{aligned} \sum_{v \in V(G_j)} deg_{G_j}(v) &= 2m(G_j) + k - 1 \\ \Rightarrow \quad k|V(G_j)| &= 2m(G_j) + k - 1, \text{ since } G \text{ is } k\text{-regular.} \\ \Rightarrow \quad k(|V(G_j)| - 1) &= 2m(G_j) - 1. \end{aligned}$$

While right hand side is an odd integer, left hand side is an even integer since $|V(G_j)|$ is odd. This contradiction proves the claim.

We next estimate the number of edges in $[S, V - S]$ in two different ways.

$$\begin{aligned} (1) \qquad |[S, V - S]| &\leq k|S|, \text{ since } G \text{ is } k\text{-regular.} \\ (2) \qquad |[S, V - S]| &= \sum_{i=1}^{t} |[S, V(G_i)]|, \text{ since } V - S = V(G_1) \cup \cdots \cup V(G_t) \\ &\geq \sum_{i=1}^{t} k, \text{ (by the above claim)} = tk. \end{aligned}$$

(1) and (2) imply that $t \leq |S|$. Hence, G satisfies (Tutte) and it has a perfect matching. □

Corollary (Petersen, 1891)**.** *If G is a 2-edge-connected 3-regular graph, then G has a perfect matching.* □

However, every connected 3-regular graph does not have a perfect matching; see Figure 8A.8. Hence, we cannot strengthen Petersen's result by assuming that G is connected and 3-regular.

8A.5 Greedy and approximation algorithms

Since finding $\beta_0(G)$ and $\alpha_0(G)$ are known to be hard problems, there have been attempts to describe algorithms which take G as an input and then output a vertex-cover (or an independent set) with $c\beta_0(G)$ (or $c\alpha_0(G)$) number of vertices

where c is a constant. Obviously, in the case of $\beta_0(G)$, $c \geq 1$, and in the case of $\alpha_0(G)$, $c \leq 1$. There is much research to describe algorithms to output vertex covers containing $c\beta_0(G)$ vertices with c as small as possible (and in the case of independent sets with c as large as possible). The constant c is called the ***approximation-factor*** and it is used to measure the efficiency of an algorithm. Obviously among the two algorithms A_1 and A_2 that output $c_1\beta_0(G)$ and $c_2\beta_0(G)$ vertices respectively, where $c_1 < c_2$, A_1 is a better algorithm.

What we have learned in the previous sections is enough for us to describe two such algorithms.

An approximate vertex-cover algorithm with approximation factor 2.

Input: A graph G on n vertices, $M \leftarrow \emptyset$, $K \leftarrow \emptyset$.

Output: A maximal matching M of G and a vertex-cover K of G.

- While $E(G) \neq \emptyset$ do
  - Select any edge $e(u, v)$ from $E(G)$.
  - $M \leftarrow M \cup \{e\}$, $K \leftarrow K \cup \{u, v\}$.
  - $G \leftarrow G - \{u, v\}$.
- end while.
- Output M and K.

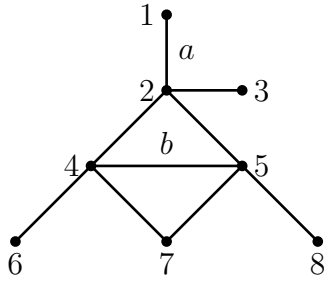
We show that $|K| \leq 2\beta_0(G)$.

Theorem 8A.10 (Correctness of the algorithm)**.** *Any output K is a vertex-cover of G with $|K| \leq 2\beta_0(G)$.*

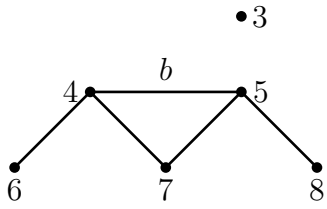
Proof. At the outset observe that M is indeed a maximal matching. Since at every step after including $e(u, v)$ in M we have deleted all the edges incident to u or v. Since M is a maximal matching, the set of all M-unsaturated vertices $V - K$ is an independent set. That is, every edge in G is incident with a vertex in K. Hence, K

is a vertex-cover such that $|K| \leq 2|M| \leq 2\alpha_1(G) \leq 2\beta_0(G)$; since $\alpha_1(G) \leq \beta_0(G)$, for any graph G (see the observation made in the beginning of the Section 7.4). □

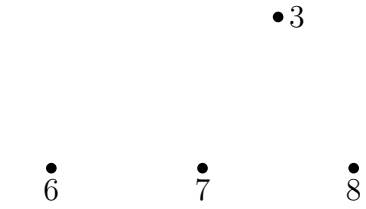
An illustration for the algorithm:



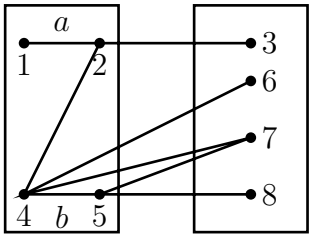
(a) G



(b) $M \leftarrow \{a\}$, $K \leftarrow \{1,2\}$, $G_1 = G - \{1,2\}$



(c) $M \leftarrow \{a,b\}$, $K \leftarrow \{1,2,4,5\}$, $G_2 = G - \{1,2\} - \{4,5\}$



$K = \{1,2,4,5\}$ $V - K$

Figure 8A.11: A graph G with a maximal matching $\{a, b\}$ generated by the greedy algorithm.

- Notice that if we had selected the edge (4,6) instead of (4,5) in the second step, we would have generated a matching with 3 edges which is also a maximal matching.

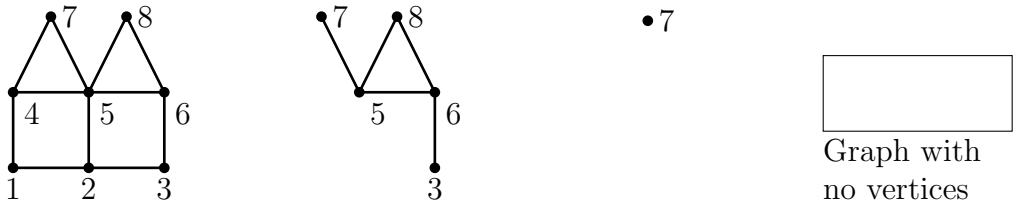
A greedy algorithm to output a maximal independent set of vertices.

Input: A graph G, $I \leftarrow \emptyset$.

Output: A maximal independent set I of vertices.

- While $V(G) \neq \emptyset$ do
  - Select any vertex (arbitrarily) say v from $V(G)$.
  - $I \leftarrow I \cup \{v\}$, $G \leftarrow G - \{v\} - N_G(v)$
- end while.
- Output I.

An illustration:



(a) G; $I \leftarrow \emptyset$ (b) $I \leftarrow \{1\}$, $G_1 = G - \{1\} - \{2, 4\}$. (c) $I \leftarrow \{1, 6\}$, $G_2 = G_1 - \{6, 3, 5, 8\}$. (d) $I \leftarrow \{1, 6, 7\}$, $G_3 = G_2 - \{7\}$.

Figure 8A.12: A graph G with maximal independent set $I = \{1, 6, 7\}$ generated by the greedy algorithm.

Remarks.

- The algorithm indeed outputs an independent set since after selecting a vertex v we delete v and all the vertices adjacent to v.
- The algorithm need not generate a maximum independent set. If we had selected the vertex 3 in G_1 instead of 6 and thereafter had selected 7, we could have generated an independent set $\{1, 3, 7, 8\}$ which is a maximum independent set.

Theorem 8A.11. *The algorithm outputs an independent set I with at least* $\dfrac{n}{1 + \Delta(G)}$ *vertices.*

Proof. The algorithm indeed outputs a maximal independent set of vertices, since at every step after selecting vertex v, we delete v and all its neighbors, and in the next step, we select a new vertex from $G - v - N(v)$. Suppose $I = \{v_1, v_2, \ldots, v_t\}$ is the output. After selecting v_t, we are left with no more vertices in the residual graph. So

$$\begin{aligned} n &\leq \underbrace{(1+\Delta) + \cdots + (1+\Delta)}_{\text{sum } t \text{ times}}, \\ &\qquad \text{since in each step we have deleted at most } 1+\Delta \text{ vertices.} \\ &= t(1+\Delta) \end{aligned}$$

$$\text{Hence,} \qquad |I| = t \geq \frac{n}{1+\Delta}. \quad \square$$